Pooling and Drift in Delayed Bandits

Melika Baghi
Georgia Institute of Technology
mbaghi3@gatech.edu

Abstract

A system often has to act long before it learns whether the act worked: a recommender sees a click in seconds and a purchase in days. With K actions and a delay of d rounds, the best rate known for this setting is $\widetilde{O}(\sqrt{(K+d)T})$ over T rounds (Esposito et al., 2023), so a longer menu is always more expensive to learn from. It need not be: if the outcome depends on the action only through the state it produced, a click or a browse or a cart, then one late outcome informs every action that could have produced the state observed, and the price is set by how many genuinely different states the actions produce rather than by how many actions there are. We measure that by an *effective dimension* v_t between 1 and the number of states, and prove $\widetilde{O}(\sqrt{(d+1)V\log K})$ for a rotating algorithm and $\widetilde{O}(\sqrt{V^-}+\sqrt{dT})$ for the single-copy one that is actually used, for any budget fixed in advance; merging similar states lowers the price again, at a bias we make explicit. Even when handed the exact losses from d rounds ago, no algorithm escapes $\Omega(\sqrt{d\mathcal{E}\min\{1+\log J, T/d\}})$, where J counts the drifting directions and $\mathcal{E}$ bounds how far the losses move while the learner waits. On generated data the state channel cuts regret by up to 79 per cent against action-level weighting, and on the funnel family by 32 to 68 per cent against the minimax-optimal method of Zimmert and Seldin (2020) under the tuning protocol we report.

1 Introduction

Delayed outcomes are common in sequential decision-making: a system must act now even though the outcome it cares about may arrive much later. In recommendation, an item is shown immediately, an engagement signal such as a click or browsing depth is observed soon after, and conversion or retention may only be known days or weeks later, so decisions are made before their economically relevant consequences appear. The central question is when such a signal can reduce the cost of learning before the final outcome arrives.

The pattern is not special to recommendation: a markdown is set and sell-through is known a season later, staff are rostered and the shift's outcome lands later, marketing spend commits before attribution resolves. Here an action is an item that could be shown, a state is what the user does next, and the delayed outcome is whether they eventually buy. Different items may induce similar distributions over engagement states, so an outcome observed after one item can carry information about many others. The difficulty scales not with the number of actions K but with how many genuinely different operational regimes those actions induce, so thousands of items funnelling users through a few engagement patterns stay easy to learn from.

The question sits between machine learning and operations research. Exploiting the channel from actions to states asks what can be inferred from data; the wait before the outcome arrives is a constraint on the operation, which no estimator removes; and the decision commits while only the first of the two has been resolved. Bandits with intermediate observations formalize this: the learner repeatedly picks one action and sees the consequence of that action alone. Esposito et al. (2023) show

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

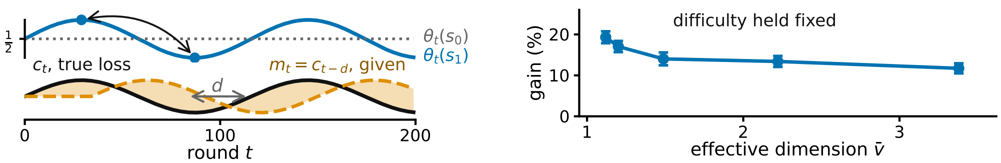

Figure 1: **An illustration of the two limits on intermediate feedback, on generated data. Left,** what each state predicts about the outcome changes while the map from actions to states stays fixed, so the losses from d rounds earlier grow increasingly inaccurate; E_2 of Section 3 tracks it. **Right,** more overlap between actions allows more sharing, at matched difficulty, $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$, 20 paired seeds, bars one standard error of the paired difference (Appendix E, study 10).

that the *state-to-loss* map decides the complexity, an unrestricted one giving only the rate available with no intermediate observations; we fix that map and ask how much the *action-to-state* structure helps. The delayed-feedback literature has since developed along other axes: Bar-On and Mansour (2025) obtain instance-dependent bounds when the observations themselves evolve, and Levy et al. (2025) treat contextual actions under delay with general function approximation. In neither does an intermediate state carry one delayed outcome across several actions, which is the mechanism studied here.

Intermediate observations face two independent limitations: whether information can be shared *across actions*, and whether it stays useful *across time*.

Across actions: because similar actions reach similar states, we can *pool through the state*, charging a delayed outcome to every action that could have produced the state observed. What this costs is governed by an *effective dimension* v_t, measuring how distinguishable the action-induced state distributions are under the current play; it can be far smaller than K.

Across time: delay creates a second difficulty even when the learner is handed an exact description of the past. Let $m_t = c_{t-d}$ be the action-loss vector from d rounds earlier. Its usefulness depends on how far the losses have moved, which we measure by $E_2 = \sum_t \|c_t - m_t\|_\infty^2$ (Figure 1). Our lower bound asks how much regret this mismatch forces even when m_t is supplied for free.

Adapting to overlap and drift at once remains open (Section 6).

2 Related work

Four ingredients separate the neighbouring settings: whether the outcome is delayed, whether a state is seen, whether the channel producing it is known, and whether the bound is instance-dependent.

	delayed	state seen	P known	instance-dependent
Zimmert and Seldin (2020)	✓	✗	✗	✗
Esposito et al. (2023)	✓	✓	✗	✗
Eldowa et al. (2024)	✗	✓	✓	✓
Bar-On and Mansour (2025)	✓	✗	✗	✓
this paper	✓	✓	✓	✓

Where this work sits: the four ingredients that separate the neighbouring settings.

The last row is the contribution: no prior work combines all four. Eldowa et al. (2024) is closest, and $v_t - 1$ is exactly the χ^2 mutual information they introduce, so that quantity is theirs; what is new is its role once the outcome is delayed (Appendix C). Vernade et al. (2020) vary the action-to-state map while fixing the state-to-loss map, where we do the reverse; Wang et al. (2026) and Zhang et al. (2026) measure a prediction error as E_2 does, without the delay; and feedback graphs (Alon et al., 2015; Esposito et al., 2022; Gabbianelli et al., 2023) reveal other actions' losses, where here the state's loss must still be estimated.

3 Problem formulation

The question is whether the intermediate state can lower the cost of learning below what K alone forces. The model that makes it precise is the following. There are T rounds, K actions, a finite state space $\mathcal{S}$, a known action-to-state matrix P, and a fixed known delay d; all logarithms are natural and $\widetilde{O}$ hides logarithmic factors. Every round follows the chain $A_t \xrightarrow{P} S_t \xrightarrow{\theta_t} X_t$: the learner draws A_t from a distribution x_t, observes $S_t \sim P(\cdot \mid A_t)$ at once, and receives the delayed outcome only at time $t+d$, encoded as a loss $X_t \in [0,1]$ so that smaller is better, with $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$, where $\mathcal{F}_{t-1}$ is everything seen before round t; outcomes are conditionally independent across rounds given the states. The conditional mean depends on A_t only through S_t. This *state-sufficiency* assumption licenses cross-action pooling: an outcome observed at state S_t may update every action that could have produced that state. If an action affects the outcome directly, in a way not represented in S_t, the argument no longer applies. Here x_t depends only on $\mathcal{F}_{t-1}$; X_r may first affect the action distribution at round $r+d+1$, equivalently the round-t decision uses outcomes only from rounds $r \le t-d-1$.

The matrix $P \in [0,1]^{K\times|\mathcal{S}|}$ is row-stochastic and fixed, and the *state-loss means* $\theta_t \in [0,1]^{|\mathcal{S}|}$ are *oblivious*: chosen in advance, not reacting to the learner. Action losses are $c_t = P\theta_t$ and performance is the static *pseudo-regret*, the gap to the best single action in hindsight, $\mathbb{E}[R_T] = \mathbb{E}[\sum_t c_t(A_t)] - \min_a \sum_t c_t(a)$. **Oracle for the lower bound.** For Theorem 2 only, the learner also receives the *stale action-loss vector* $m_t \triangleq c_{t-d}$ before choosing A_t, with $m_t = c_1$ for $t \le d$. This is not a function of the observables, so a lower bound surviving it is stronger, and STATE-EXP3 never uses it. We measure how misleading m_t is by $E_2 = \sum_t \|c_t - m_t\|_\infty^2 \le T$, which stays small when errors are small even if frequent. The geometry of P governs sharing across actions and the evolution of θ_t governs staleness across time.

4 What the theory guarantees

Two results follow: the number of actions stops setting the price for the algorithm one would actually run, and no sharing removes the cost of waiting. A rotating variant and one for merged states are in Appendix B.

An arriving outcome need not update the action played alone. Every action can be credited by how likely it was to have produced the observed state. Here P is known and m_t is not given. For each state $s \in \mathcal{S}$ let $q_t(s) = \sum_{a=1}^K x_t(a)P(s \mid a)$, and give each action $a \in \{1,\dots,K\}$ the estimate $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$. It is correct on average, and its noise is governed by the *effective dimension* $v_t = \sum_s q_t(s)^{-1}\sum_a x_t(a)P(s \mid a)^2 \in [1,|\mathcal{S}|]$. STATE-EXP3 runs m copies of EXP3 (Auer et al., 2002) in rotation at learning rate η, copy $i \in \{0,\dots,m-1\}$ taking every mth round, and adds $\hat{c}_t$ to a copy's running total once the outcome is usable. A round costs $O(K)$ (Appendix A, with pseudocode and the χ^2 reading of v_t).

Theorem 1. *Let P be known and (θ_t) oblivious, and fix in advance any V^- with $\sum_t(v_t-1) \le V^-$ always. Then* STATE-EXP3 *with $m = 1$ and any $\eta \le 1/(e(d+1))$ satisfies $\mathbb{E}[R_T] \le \sqrt{2(3.3V^- + 2dT)\log K} + e(d+1)\log K + d$.*

So K is replaced by V^-, which counts how differently the actions behave rather than how many there are, and the delay enters additively at $\widetilde{O}(\sqrt{V^-} + \sqrt{dT})$, for the algorithm every experiment below runs. Writing $\bar{v}$ for the largest v_t over all ways of playing, $V^- = (\bar{v}-1)T$ is admissible and depends on P alone. Rotating $d+1$ copies and merging similar states are handled in Appendix B; none of the three adapts to E_2.

The proof turns on one pathwise identity. The denominator of $\hat{c}_t$ is the state marginal of the distribution supplying its numerator, so $\langle x_t, \hat{c}_t\rangle = X_t \le 1$ on every path, whatever $q_t(S_t)$ is. That bounds how far the play distribution moves while an outcome is in flight, which is what the delayed cross-term needs; centring by $X_t\mathbf{1}$ replaces v_t by $v_t - 1$ (Appendix F).

Where this stops. How much does waiting cost even when the stale losses are free? Theorem 2 shows a cost from time alone, not a joint overlap-drift bound: the construction gives each of the J drifting directions a private state, removing useful pooling and forcing $J \le |\mathcal{S}|-1$ (Appendix D).

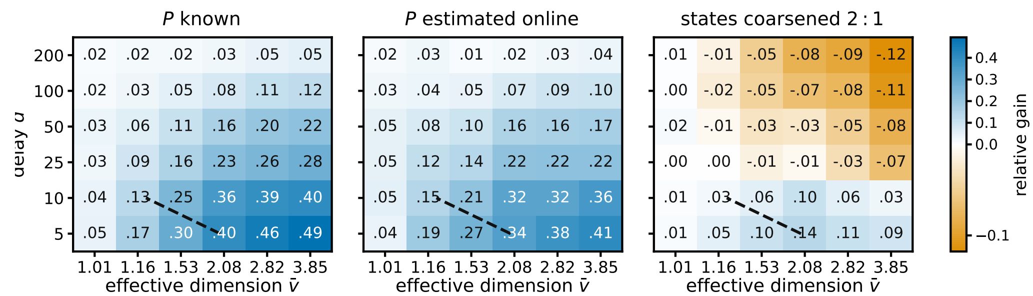

Figure 2: **Where pooling pays.** The gain should follow the effective dimension rather than the number of actions, and should survive when P is estimated rather than known. Both hold across the sweep. Relative gain $(R_{\text{action}} - R_{\text{state}})/R_{\text{action}}$, single copy, $K = 40$, $|\mathcal{S}| = 8$, $T = 5000$, 12 paired seeds; blue is a gain. $\widehat{v}$ estimates $\sup_x v(x)$ for the raw map by Monte Carlo, which for the coarsened panel is not the dimension of the coarsened map that the grouping bound charges. The dashed line compares two upper bounds for the rotating variant, so it is a reference rather than a prediction for the variant plotted.

Theorem 2. *Fix T, a delay $1 \le d \le T/2$, a drift budget $\mathcal{E} \le T/4$, and $1 \le J \le \min\{K-1, |\mathcal{S}|-1\}$. Every algorithm, even one handed the exact losses from d rounds ago, meets some instance fixed in advance with $E_2 \le \mathcal{E}$ on which $\mathbb{E}[R_T] = \Omega(\sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$. Without that help, and if $K \le T$, the best regret any algorithm can guarantee is $\Omega(\sqrt{KT} + \sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$, up to constants.*

The first term is the usual cost of exploring. The second is the cost of delay: it grows with the wait, how far the losses move during it, and the number J of directions the best fixed action can use. They come from different hard instances, so adding them overstates the truth by at most a factor of two.

5 Experiments

A claim about instances is settled where the two counts it separates, how many actions there are and how many distinct behaviours they have, come apart. The families below are built so they do. With $K = 200$ generated items and six levels of engagement the estimated dimension is 1.97, and the state-pooled variant cuts regret against action-level weighting by 79.3, 63.6 and 38.9 per cent at $d = 10, 50, 200$. That setting is hand-structured, not fitted to data, so it shows the mechanism, not a deployed system.

It also beats the minimax-optimal delayed-bandit method of Zimmert and Seldin (2020), tuned over a grid under the reported tuning protocol.

d	action-level	tuned (Zimmert and Seldin, 2020)	State-EXP3	cut	cut
10	4577	2959	949	79.3%	67.9%
50	4752	3557	1728	63.6%	51.4%
200	5307	4735	3245	38.9%	31.5%

Funnel family, $K = 200$, $|\mathcal{S}| = 6$, $T = 20000$, 8 paired seeds; cuts are against action-level weighting and against the tuned baseline (study 16).

These runs use the single-copy algorithm, the one Theorem 1 covers; Appendix E also reports the rotating variant. A best-state rule leads seven of nine settings, but where its assumption fails its regret grows linearly and its spread across seeds is four times ours.

Figure 2 sweeps row overlap against delay and marks where pooling pays, in the three settings the studies treat separately: P known, P estimated, and states coarsened.

6 Discussion

Pooling helps when many actions lead to similar states, and that is checkable before deployment, since P is known: an estimate of $\bar{v}$ well below K indicates the state channel is worth using.

Theorem 2 says what no algorithm can do about the second: the uncertainty left by waiting is irreducible, surviving even when the exact losses from d rounds ago are handed over free. Two limits remain: $\bar{v}$ is estimated rather than solved, and the bounds lie on different axes.

References

N. Alon, N. Cesa-Bianchi, O. Dekel, and T. Koren. Online learning with feedback graphs: beyond bandits. *COLT*, PMLR 40, 2015.

P. Auer, N. Cesa-Bianchi, Y. Freund, and R. E. Schapire. The nonstochastic multiarmed bandit problem. *SIAM Journal on Computing*, 32(1):48–77, 2002.

Y. Bar-On and Y. Mansour. Non-stochastic bandits with evolving observations. *ALT*, PMLR 272, 2025.

N. Cesa-Bianchi and G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.

N. Cesa-Bianchi, C. Gentile, and Y. Mansour. Delay and cooperation in nonstochastic bandits. *JMLR*, 20, 2019.

K. Eldowa, N. Cesa-Bianchi, A. M. Metelli, and M. Restelli. Information capacity regret bounds for bandits with mediator feedback. *JMLR*, 25(353), 2024.

E. Esposito, S. Masoudian, H. Qiu, D. van der Hoeven, N. Cesa-Bianchi, and Y. Seldin. Delayed bandits: When do intermediate observations help? *ICML*, PMLR 202, 2023.

E. Esposito, F. Fusco, D. van der Hoeven, and N. Cesa-Bianchi. Learning on the edge: online learning with stochastic feedback graphs. *NeurIPS*, 35, 2022.

G. Gabbianelli, G. Neu, and M. Papini. Online learning with off-policy feedback. *ALT*, PMLR 201, 2023.

O. Levy, L. Erez, A. Cohen, and Y. Mansour. Regret bounds for adversarial contextual bandits with general function approximation and delayed feedback. *arXiv:2510.09127*, 2025.

T. S. Thune, N. Cesa-Bianchi, and Y. Seldin. Nonstochastic multiarmed bandits with unrestricted delays. *NeurIPS*, 32, 2019.

C. Vernade, A. György, and T. Mann. Non-stationary delayed bandits with intermediate observations. *ICML*, PMLR 119, 2020.

S. Wang, A. Barik, and V. Y. F. Tan. Bandit convex optimization with gradient prediction adaptivity. *arXiv:2605.22191*, 2026.

K. W. Zhang, T. Baldwin-McDonald, K. Ciosek, L. Maystre, and D. Russo. Impatient bandits: optimizing for the long-term without delay. *JMLR*, to appear, 2026.

J. Zimmert and Y. Seldin. An optimal algorithm for adversarial bandits with arbitrary delays. *AISTATS*, PMLR 108, 2020.

A The state-pooled estimator

The algorithm. Inputs are the action-to-state matrix P, the delay d, the number of copies m, and the rate $\eta > 0$. The interleaved setting is $m = d + 1$ and the single-copy one is $m = 1$; Theorems 5 and 1 cover them respectively. Round r's outcome becomes usable at round $r + d + 1$, which for $m = d + 1$ is the next round belonging to the copy that played round r.

1. Set $L_i \leftarrow (0,\dots,0) \in \mathbb{R}^K$ for $i = 0,\dots,m-1$, and let $\mathcal{P}$ be an empty table of rounds awaiting their outcome.
2. For $t = 1,\dots,T$:
   (a) *Deliver.* If $r \triangleq t-d-1 \geq 1$, read $(i_r, q_r(S_r), S_r, X_r)$ from $\mathcal{P}$, remove it, and update only that copy, $L_{i_r}(a) \leftarrow L_{i_r}(a) + P(S_r \mid a)X_r/q_r(S_r)$ for every a. This is the pooled step, and it touches every action rather than the one played.
   (b) *Select the copy.* $i \leftarrow t \bmod m$.
   (c) *Play.* $x_t(a) \propto \exp(-\eta L_i(a))$, draw $A_t \sim x_t$, and observe S_t at once.
   (d) *Record.* Compute the scalar $q_t(S_t) = \sum_a x_t(a)P(S_t \mid a)$ and store $(i, q_t(S_t), S_t, \cdot)$ in $\mathcal{P}$, to be completed when X_t arrives at time $t+d$.

Steps (a), (c) and (d) are each $O(K)$ and there is no optimization subproblem, so a round costs $O(K)$ once P is stored. The state is held in the m vectors L_i, which is mK numbers, the matrix P, which is $K|\mathcal{S}|$ numbers, and four scalars for each of the at most d rounds in flight. Only the scalar $q_r(S_r)$ is needed from round r, not the vector x_r.

A worked example. Take three actions and two states, with

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \\ 0 & 1 \end{pmatrix}, \qquad x_t = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}),$$

so $q_t(s_1) = \frac{1}{2}(0.8) + \frac{1}{4}(0.4) + \frac{1}{4}(0) = 0.5$ and $q_t(s_2) = 0.5$. The third action cannot produce s_1 at all. Suppose the learner draws a_1, observes s_1, and receives $X_t = 1$ at time $t+d$. Then $\hat{c}_t(a) = P(s_1 \mid a)/0.5$, so $\hat{c}_t = (1.6, 0.8, 0)$ against the action-level $\tilde{c}_t = (2,0,0)$. One observation updates all three actions: a_2 is charged although it was never played, because it had a 0.4 chance of producing the state seen, and a_3 is charged nothing, because it could not have produced that state. Nothing is biased by this. With $\theta_t = (1,0)$ the true losses are $c_t = (0.8, 0.4, 0)$, and averaging $\hat{c}_t$ over $S_t \sim q_t$ returns exactly that. What is bought is the second moment: $v_t = 1.44$ here, against $K = 3$ for the action-level estimate, and with $\theta_t = (1,1)$ both are attained exactly.

Remark 3 (The χ^2 form). Under x_t the pair (A_t, S_t) has law $x_t(a)P(s \mid a)$ with marginals x_t and q_t, so

$$\begin{aligned}\chi^2\big(P_{A_t,S_t} \,\big\|\, P_{A_t} \otimes P_{S_t}\big) &= \sum_{a,s} \frac{\big(x_t(a)P(s \mid a) - x_t(a)q_t(s)\big)^2}{x_t(a)q_t(s)} \\ &= \sum_{a,s} \frac{x_t(a)P(s \mid a)^2}{q_t(s)} - 2 + 1 \;=\; v_t - 1,\end{aligned}$$

the cross term summing to $\sum_{a,s} x_t(a)P(s \mid a) = 1$ and the last to $\sum_{a,s} x_t(a)q_t(s) = 1$. So $v_t = 1$ exactly when A_t and S_t are independent, which pins the rows of P only on the support of x_t and therefore means all rows agree whenever x_t has full support, as exponential weights does. If P is deterministic and every state is reached by some action in the support of x_t, then v_t equals the number of states that support reaches. In particular, under full-support play and an onto deterministic P, $v_t = |\mathcal{S}|$. This is an identity, not a bound; it is used for reading v_t, and once in the proof of Theorem 1.

Proposition 4. *Fix t, let x_t be the play distribution, $q_t(s) = \sum_a x_t(a)P(s \mid a)$ the induced state distribution, and $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$, which is well defined because only states with $q_t(s) > 0$ occur. For statements involving $1/x_t(a)$, restrict to actions with $x_t(a) > 0$;* STATE-EXP3 *itself has full support on every action. Then $\hat{c}_t(a) \geq 0$, $\hat{c}_t(a) \leq 1/x_t(a)$, $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$, and*

$$\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \;\leq\; v_t := \sum_{s:q_t(s)>0} \frac{\sum_a x_t(a)P(s \mid a)^2}{q_t(s)}, \qquad 1 \leq v_t \leq |\mathcal{S}|,$$

states of zero probability being omitted throughout, which is the convention $0/0 = 0$. The action-level $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$ has the same first three properties and weighted second moment at most K.

Proof. Nonnegativity is immediate from $X_t \geq 0$. Given $\mathcal{F}_{t-1}$ the state S_t has distribution q_t, and $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, S_t = s] = \theta_t(s)$ by the model, since that conditional mean does not depend on the action, so $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = \sum_s q_t(s)P(s \mid a)\theta_t(s)/q_t(s) = c_t(a)$. For the range, $q_t(s) \geq x_t(a)P(s \mid a)$ for every a, so $P(s \mid a)/q_t(s) \leq 1/x_t(a)$ and $X_t \leq 1$. For the second moment, $X_t^2 \leq 1$ gives $\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq \sum_s P(s \mid a)^2/q_t(s)$, and exchanging the sums gives v_t. Each summand of v_t is at most 1, since $P(s \mid a) \leq 1$ makes $\sum_a x_t(a)P(s \mid a)^2 \leq q_t(s)$, and at least $q_t(s)$, since Jensen applied to $a \mapsto P(s \mid a)$ under x_t gives $\sum_a x_t(a)P(s \mid a)^2 \geq q_t(s)^2$. Summing the two bounds gives $1 \leq v_t \leq |\mathcal{S}|$. The action-level claims are the same computation with the state replaced by the played action, whose weighted second moment is $\sum_a x_t(a)/x_t(a) = K$. $\square$

The two ends are attained. If every action induces the same state distribution then $v_t = \sum_s q_t(s) = 1$, and if P is deterministic, each action reaching a single state, then every $P(s \mid a)$ is 0 or 1 and each summand is exactly 1, so v_t counts the states reached and equals $|\mathcal{S}|$ when every state has an action reaching it. Disjoint row supports give $v_t = K$ rather than $|\mathcal{S}|$, since the summands over one action's private states add to $\sum_s P(s \mid a) = 1$, so a sensor with many private states per action resolves few of them. So v_t reads the overlap between the rows of P rather than their length, and $\bar{v} = \sup_x v(x)$, the supremum over play distributions, depends on P alone. An outcome observed at a state updates the estimate for every action that can reach it, which is why the construction of Section 4 gives each drifting coordinate a private state, forcing $v_t = |\mathcal{S}|$ there. The immediately observed pairs (A_t, S_t) identify P separately, though Theorem 5 assumes it known.

B Proofs: pooling and grouping

Two results are stated here rather than in the body, since Theorem 1 covers the algorithm actually run. The first is the interleaved variant.

Theorem 5. *Let P be known and (θ_t) oblivious. Pick before the run any fixed number V with $\sum_t v_t \leq V$ always. Then* STATE-EXP3 *with $m = d + 1$ and the matching learning rate satisfies $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V \log K}$. Writing $\bar{v}$ for the largest v_t over all ways of playing, $V = \bar{v}T \leq |\mathcal{S}|T$ is one such choice, and it depends on P alone.*

The second merges states that are not worth telling apart. Group them with a map g from $\mathcal{S}$ onto a smaller set $\mathcal{Z}$ and run the same algorithm on $g(S_t)$, where $P^g(z \mid a) = \sum_{s:g(s)=z} P(s \mid a)$. Grouping never raises the dimension, but the estimate now tracks a group average, so the widest spread of losses inside a group, $\delta_t(g) = \max_z \max_{s,s':g(s)=g(s')=z} |\theta_t(s) - \theta_t(s')|$, is the error introduced.

Theorem 6. *Under the hypotheses of Theorem 5, running on g with $\sum_t v_t(g) \leq V(g)$ gives $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V(g) \log K} + 2\sum_t \delta_t(g)$.*

Merging need not preserve state-sufficiency, so the grouped estimate is right not for c_t but for a stand-in loss differing from it by at most $\delta_t(g)$ on every action. Keeping states apart preserves differences that matter but learns more slowly; merging shares more but pays $2\sum_t \delta_t(g)$. The best grouping is the true one: with 16 states from four hidden groups, sizes $1, 2, 4, 8, 16$ give regret 1590, 831, 754, 920, 1080, lowest at four, which no algorithm was told. The map g is fixed in advance; learning it from data remains open.

Proof idea. The pooled estimator is unbiased for every action, while Proposition 4 bounds its play-weighted second moment by v_t rather than K. With $m = d + 1$ interleaved copies, the outcome generated on a round belonging to one copy is available before that copy acts again, so each copy is an ordinary exponential-weights process with immediate feedback. Summing the $d + 1$ potential inequalities gives $(d + 1) \log K/\eta$, and the pooled second moments give $\frac{\eta}{2} \sum_t v_t$.

Proof of Theorem 5. Fix a copy i and let T_i be the number of rounds it owns, so $\sum_i T_i = T$. Round t's outcome is usable from round $t + d + 1$, which with $m = d + 1$ is exactly $t + m$, the copy's next round, so within a copy the feedback is undelayed and L_i carries exactly the estimates of that copy's earlier rounds. Exponential weights on nonnegative losses give, for any action $a^\star$ and any $\eta > 0$,

$$\sum_{t \in i} \big(\langle x_t, \hat{c}_t \rangle - \hat{c}_t(a^\star)\big) \leq \frac{\log K}{\eta} + \frac{\eta}{2} \sum_{t \in i} \sum_a x_t(a)\hat{c}_t(a)^2,$$

by the usual potential telescoping with $e^{-z} \le 1 - z + z^2/2$, which needs $z \ge 0$ and therefore $\hat{c}_t \ge 0$, supplied by Proposition 4. Each x_t is $\mathcal{F}_{t-1}$-measurable, because L_i then holds only rounds $r \le t - m = t - d - 1$, so taking expectations turns the left side into $\sum_{t \in i}(\langle x_t, c_t\rangle - c_t(a^\star))$ by unbiasedness and the quadratic term into at most $\mathbb{E}[\sum_{t\in i} v_t]$, again by Proposition 4. Summing over the $m = d{+}1$ copies and using $\sum_i \min_a \sum_{t\in i} c_t(a) \le \min_a \sum_t c_t(a)$, which is what lets the copies be compared against one common action, gives $\mathbb{E}[R_T] \le (d+1)\log K/\eta + \frac{\eta}{2}\mathbb{E}[\sum_t v_t]$. If $\sum_t v_t \le V$ surely then $\eta = \sqrt{2(d+1)\log K/V}$ balances the two terms and gives $\sqrt{2(d+1)V\log K}$, and $V = \bar{v}T$ is admissible by the definition of $\bar{v}$. □

The delay enters as a factor because the copies do not share their estimates, each paying $\log K/\eta$ over $T/(d+1)$ rounds. Sharing them is exactly the $m = 1$ algorithm of the open problem of Section 6.

Lemma 7 (Coarsening)**.** *For every play distribution and every g, $v_t(g) \le v_t$.*

Proof. It suffices to merge two states, the general case following by induction on the merges, and a merged group of zero probability contributes nothing on either side by the convention above. Write P_1, P_2 for the two columns, $A_j = \sum_a x_t(a)P_j(a)^2$, $B_j = \sum_a x_t(a)P_j(a)$ and $\lambda = B_1/(B_1+B_2)$. By Cauchy–Schwarz, $(P_1{+}P_2)^2 \le \lambda^{-1}P_1^2 + (1{-}\lambda)^{-1}P_2^2$ pointwise in a, so $\sum_a x_t(a)(P_1{+}P_2)^2 \le (B_1+B_2)(A_1/B_1 + A_2/B_2)$, and dividing by the merged denominator $B_1 + B_2$ shows the merged summand is at most $A_1/B_1 + A_2/B_2$, the two it replaces. □

Proof of Theorem 6. Running the algorithm on g is running it on the sensor $(\mathcal{Z}, P^g)$. The surrogate losses below depend on x_t and so are predictable rather than oblivious, but the proof of Theorem 5 uses obliviousness nowhere, only that each c_t^g is $\mathcal{F}_{t-1}$-measurable given the play, so it applies unchanged and bounds regret measured in the surrogate losses $c_t^g(a) = \sum_z P^g(z \mid a)\bar{\theta}_t(z)$, where $\bar{\theta}_t(z) = \sum_{s:g(s)=z} q_t(s)\theta_t(s)/q_t^g(z)$ is the group mean the grouped estimator is unbiased for, by the same computation as in Proposition 4 with S_t replaced by $g(S_t)$. Fix a and z. Both $\{q_t(s)/q_t^g(z)\}$ and $\{P(s \mid a)/P^g(z \mid a)\}$ are probability vectors on $\{s : g(s) = z\}$, so their θ_t-averages lie in a common interval of length $\delta_t(g)$ and differ by at most $\delta_t(g)$. Weighting by $P^g(z \mid a)$ and summing over z gives $|c_t^g(a) - c_t(a)| \le \delta_t(g)$ for every a, hence $\langle x_t, c_t\rangle - c_t(a^\star) \le \langle x_t, c_t^g\rangle - c_t^g(a^\star) + 2\delta_t(g)$. Summing over t adds $2\sum_t \delta_t(g)$, and Lemma 7 makes $V(g)$ no larger than a bound valid for the raw states. □

C Relation to other inaccuracy measures

Write $\Lambda_2 = \sum_t \min\{1, \|c_t - m_t\|_2^2\}$ for the inaccuracy measure of Bar-On and Mansour (2025). Unlike E_2, which charges only the worst action in a round, Λ_2 grows with how many actions are predicted wrongly, and since $\|v\|_\infty \le \|v\|_2 \le \sqrt{J}\|v\|_\infty$ for a vector v with J nonzero coordinates,

$$E_2 \;\le\; \Lambda_2 \;\le\; J\,E_2,$$

with both ends attained: the left when a single action is mispredicted, the right when all J are mispredicted equally. So a bound in E_2 is never weaker and can be J times stronger, which is why Theorem 2 is stated in E_2. The two measures agree exactly when the prediction error is concentrated on one action.

D Proofs: the lower bound

The measure E_2 is controlled by the state-loss variation W, and no converse bound holds in general.

Lemma 8 (Delay window)**.** *$E_2 \le d^2 W$ for $W = \sum_t \|\theta_t - \theta_{t-1}\|_\infty^2$. The ratio $E_2/(d^2W)$ tends to one when θ drifts monotonically in one coordinate at a constant rate and $T/d \to \infty$, the first d rounds contributing $\sum_{j<d} j^2$ rather than d^2 each under the convention $m_t = c_1$.*

In words, accumulated squared prediction error is at most the state-loss variation inflated by the square of the delay, and no better bound in W alone is available.

Proof of Lemma 8. Throughout write $\Delta_j = \theta_j - \theta_{j-1}$, set $\theta_0 = \theta_1$, so $\Delta_1 = 0$, and read c_r as c_1 for $r \le 0$, matching the convention $m_t = c_1$ for $t \le d$ of Section 3. Then $c_t - c_{t-d} = P \sum_{j=t-d+1}^{t} \Delta_j$. Because the rows of P are probability vectors, averaging cannot increase the supremum norm, so $\|c_t - c_{t-d}\|_\infty \le \sum_j \|\Delta_j\|_\infty$. By Cauchy–Schwarz applied to the d summands, $\|c_t - c_{t-d}\|_\infty^2 \le d \sum_{j=t-d+1}^{t} \|\Delta_j\|_\infty^2$. Summing over t and using that each increment belongs to at most d windows gives $E_2 \le d^2 W$. The ratio $E_2/(d^2 W)$ approaches one when every increment has the same magnitude, sign and active coordinate, and some action reaches that coordinate deterministically, since then no cancellation occurs within a full window and P passes the increment through undiminished. Without the second condition the drift can lie in the kernel of P and the ratio can be zero. □

Lemma 9 (Information isolation)**.** *If $h \le d$ then A_t is independent of the signs governing its own block, and this remains true when the learner is provided with the exact stale action-loss vector m_t.*

Proof of Lemma 9. For t in block b, every realized outcome usable when choosing A_t comes from a round $r \le t - d - 1$. Since $t \le bh$ and $h \le d$, we have $r \le bh - d - 1 < (b-1)h$, so every usable realized outcome depends only on signs from blocks strictly before b, whose joint law depends only on $\sigma_2, \dots, \sigma_{b-1}$. Hence by conditional independence A_t is independent of block b's signs. For the oracle, take any $h \le d$. If $t > d$ then $t - d \le bh - d \le (b-1)h$ by the same computation, so $m_t = c_{t-d}$ is measurable with respect to the signs of blocks strictly before b. If $t \le d$ then $m_t = c_1 = \frac{1}{2}\mathbf{1}$, which is deterministic because block 1 is neutral. In both cases m_t is independent of block b's signs, so adjoining it to the learner's information leaves the conclusion intact. □

The construction in full. Take states $s_0, \dots, s_q$, a stationary deterministic map sending a_j to s_j for $j \le J$ and every other action to s_0, and independent Rademacher signs $\sigma_b^{(j)}$ for $2 \le b \le B = \lfloor T/h \rfloor$. Set $\theta_t(s_0) = \frac{1}{2}$ throughout, $\theta_t \equiv \frac{1}{2}$ on block 1, and $\theta_t(s_j) = \frac{1}{2} - \varepsilon \sigma_b^{(j)}$ on each later block b. The neutral first block makes $c_1 = \frac{1}{2}\mathbf{1}$, so the convention $m_t = c_1$ for $t \le d$ of Section 3 supplies a constant over exactly the rounds that precede any usable feedback, which is what Lemma 9 needs at $b = 1$. Rounds after the last complete block carry no drift, costing a constant factor.

Lemma 10 (Rademacher lower tail)**.** *Let S_B be a sum of $B \ge 32$ independent Rademacher signs. For every $1 \le u \le \sqrt{B}/32$, $\Pr(S_B \ge u\sqrt{B}) \ge \frac{u}{4} e^{-64u^2}$.*

Proof. Write $S_B = 2X - B$ with $X \sim \mathrm{Bin}(B, \frac{1}{2})$, so that $\{S_B \ge u\sqrt{B}\} = \{X \ge B/2 + u\sqrt{B}/2\}$. Passing to X removes the lattice gap, since X takes every integer value in $[0, B]$ whereas S_B takes only those of the parity of B. Put $n = \lfloor B/2 \rfloor$ and $t = \lceil u\sqrt{B}/2 \rceil + 1$, so $X \ge n + t$ implies the event, and $t \le u\sqrt{B}$ because $u\sqrt{B} \ge \sqrt{32} > 4$.

For $j \ge 1$ the ratio of consecutive binomial probabilities is $\Pr(X = n + j)/\Pr(X = n + j - 1) = (B - n - j + 1)/(n + j)$. Using $(B-1)/2 \le n \le B/2$, its distance from one is at most $(2j-1)/(n+j) \le (4j-2)/(B-1) \le 8j/B$ for $B \ge 2$. Restricting to $j \le B/16$ keeps $8j/B \le \frac{1}{2}$, where $\log(1-x) \ge -2x$ holds, so summing over $i \le j$ gives $\Pr(X = n+j) \ge \Pr(X = n) e^{-16j(j+1)/(2B)} \ge \Pr(X = n) e^{-16j^2/B}$. The central term satisfies $\Pr(X = n) \ge 1/(2\sqrt{B})$ for every $B \ge 2$.

Sum over the t integers $j \in \{t, \dots, 2t-1\}$, all of which are at most $2t$ and hence at most $B/16$ because $u \le \sqrt{B}/32$ forces $2t \le 2u\sqrt{B} \le B/16$. Each term is at least $e^{-64t^2/B}/(2\sqrt{B})$, and $t \le u\sqrt{B}$ makes the exponent at most $64u^2$, so $\Pr(X \ge n + t) \ge t\, e^{-64u^2}/(2\sqrt{B}) \ge \frac{u}{4} e^{-64u^2}$ using $t \ge u\sqrt{B}/2$. □

Three regimes give the maximal inequality the lower bound needs. Each $S_B^{(j)}$ is sub-Gaussian with variance proxy B, so $\mathbb{E} \max_j (S_B^{(j)})^+ \le \sqrt{2B \log J} + \sqrt{B}$ throughout. For the matching lower bound, first suppose $\log J \le B/8$ and $\log J \ge 128$. Taking $u = \sqrt{\log J/128}$, which then lies in $[1, \sqrt{B}/32]$, makes $e^{-64u^2} = J^{-1/2}$, so $\Pr(S_B^{(j)} \ge u\sqrt{B}) \ge \frac{u}{4} J^{-1/2}$ and independence across j gives $\Pr(\max_j S_B^{(j)} < u\sqrt{B}) \le e^{-u\sqrt{J}/4}$, at most $\frac{1}{2}$ beyond an absolute J. Hence $\mathbb{E} \max_j (S_B^{(j)})^+ \ge \frac{1}{2} u\sqrt{B} = \Omega(\sqrt{B \log J})$. Second, if $\log J > B/8$ and $B \ge 1024$, take

$u = \sqrt{B}/32$, so $u\sqrt{B} = B/32$ and $e^{-64u^2} = e^{-B/16}$, and $J \geq e^{B/8}$ makes $J\,e^{-B/16}$ diverge, giving $\mathbb{E}\max_j (S_B^{(j)})^+ = \Omega(B)$, which is $\Omega(\sqrt{B\min\{1+\log J, B\}})$ in that regime. Third, when $\log J < 128$ or $B < 1024$ the quantity $\sqrt{B\min\{1+\log J, B\}}$ is $O(\sqrt{B})$, and a single walk already gives $\mathbb{E}(S_B^{(1)})^+ = \frac{1}{2}\mathbb{E}|S_B| \geq \frac{1}{2}(\mathbb{E}S_B^2)^{3/2}(\mathbb{E}S_B^4)^{-1/2} \geq \frac{1}{2}\sqrt{B/3}$ by Hölder, since $\mathbb{E}S_B^2 = B$ and $\mathbb{E}S_B^4 \leq 3B^2$. Combining the three regimes, $\mathbb{E}\max_j (S_B^{(j)})^+ = \Theta(\sqrt{B\min\{1+\log J, B\}})$. This is the route by which maxima of independent random walks fix the minimax rate for prediction with expert advice (Cesa-Bianchi and Lugosi, 2006).

Proof idea. Divide time into blocks of length d and give J actions private states whose losses are perturbed by independent Rademacher signs within each block. Every observation available when an action is chosen comes from an earlier block, and the supplied $m_t = c_{t-d}$ depends only on earlier signs, so the learner cannot predict the signs governing its current block. Its expected loss is therefore $T/2$, while the best fixed action in hindsight benefits from the maximum of J independent random walks. Choosing the amplitude to make $E_2 \leq \mathcal{E}$ gives the stated scale.

Proof of Theorem 2. Take $h = d$ and $\varepsilon = \frac{1}{2}\sqrt{\mathcal{E}/T}$. By Lemma 9 every algorithm's expected loss equals $T/2$, since its action is independent of the current block's signs and all states have mean $\frac{1}{2}$ marginally. Its regret is therefore exactly the comparator's fluctuation advantage, $\varepsilon d\,\mathbb{E}\big[\max_{j\leq J}\big(\sum_{2\leq b\leq B}\sigma_b^{(j)}\big)^+\big]$ with $B = \lfloor T/d \rfloor$, over the $B-1$ randomized blocks. Per round $\|c_t - m_t\|_\infty \leq 2\varepsilon$, because the supremum norm charges only the largest coordinate and each coordinate moves by at most 2ε at a block boundary; the gap is ε across the first randomized block, where m_t is still the neutral c_1, and lies in $\{0, 2\varepsilon\}$ thereafter. Hence $E_2 \leq 4\varepsilon^2 T = \mathcal{E}$ holds deterministically. The budget must bind on the realized sequence, since a budget holding only in expectation would not constrain the instance selected by Yao's principle. By Lemma 10 and the discussion following it, the expected maximum of the positive parts of J independent Rademacher sums of length $B-1$ is $\Theta(\sqrt{(B-1)\min\{1+\log J, B-1\}})$, which covers both regimes and the case $J = 1$. The hypothesis $d \leq T/2$ gives $B \geq 2$ and hence $B-1 \geq B/2$, so replacing $B-1$ by B costs a factor at most $\sqrt{2}$ in each branch of the minimum, including the saturated branch where the minimum is $B-1$ rather than $1+\log J$. Substituting $B = \lfloor T/d \rfloor$ and ε therefore gives $\Theta(\sqrt{d\mathcal{E}\min\{1+\log J, T/d\}})$. Rounds after the last complete block carry no drift and change the constant only. When the oracle is withheld, the $\sqrt{KT}$ term is Esposito et al. (2023, Thm. 5.1), whose construction is stationary and therefore lies in the present model class. ☐

Proposition 11. *Suppose $X_t = c_t(A_t)$. On instances whose drift is carried by a single action, $\Lambda_2 = E_2$, so the bound of Bar-On and Mansour (2025) reads $\widetilde{O}(\sqrt{KT + d\,E_2})$ and its drift contribution $\sqrt{d\,E_2}$ matches Theorem 2 up to logarithmic factors.*

In words, single-action drift makes the Euclidean and supremum measures agree, so on that family the drift term of the published upper bound is unimprovable in order. Its $\sqrt{KT}$ term is not matched because the oracle supplies an entire stale loss vector, so the oracle-assisted problem need not retain bandit exploration complexity. Neither result provides an E_2-adaptive upper bound under noisy outcomes.

Proof of Proposition 11. If only one coordinate of $c_t - m_t$ is nonzero then $\|c_t - m_t\|_2 = \|c_t - m_t\|_\infty$. On the family of Section 4 that coordinate has magnitude at most $2\varepsilon \leq 1$, so the truncation at one in Λ_2 is inactive and $\Lambda_2 = E_2$. The bound of Bar-On and Mansour (2025) then reads $\widetilde{O}(\sqrt{KT + dE_2})$, whose $\sqrt{dE_2}$ contribution Theorem 2 matches up to the logarithmic factor at $J = 1$. Its $\sqrt{KT}$ contribution is not matched. ☐

E Numerical detail

Sixteen studies were run and the seven bearing on the claim are reported here, each with an observation, an interpretation and a limitation. They keep their original numbers, which index the released scripts, so the numbering is not contiguous; the code for all sixteen is available. Studies 1 to 13 draw P from a Dirichlet and 14 to 16 do not. STATE-EXP3 is beaten in four of six cells of study 16 when the baseline alone is grid-tuned, and study 10 is a control for study 6 that reverses its conclusion. All policies run on the same environment parameters and common random-number streams, so a seed fixes everything except which estimator is formed, and differences are taken within a seed before averaging. Round r's outcome is consumed before round $r+d+1$, matching Section 3. Reported $\pm$ is one standard error of the paired difference. Throughout this appendix, *sample-path pseudo-regret* means $\sum_t c_t(A_t) - \min_a \sum_t c_t(a)$, evaluated on one realization of the learner and environment randomness. Averaging it over seeds estimates the expected pseudo-regret that the theorems bound. We use that name rather than "realized regret" to distinguish it from regret computed from the sampled outcomes X_t, which is not what these studies report.

Study 1. The bound holds and pooling helps. With $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, drift band 0.05 and 100 seeds:

K	state, $m = d+1$	state, $m = 1$	action, $m = 1$	paired gain	state better in
8	455.1	368.2	387.6	19.3 ± 3.4	73/100
20	618.3	493.7	547.5	53.8 ± 5.6	87/100
40	734.3	574.5	696.1	121.5 ± 9.4	99/100
80	780.4	614.8	784.0	169.3 ± 9.9	100/100

Study 1, $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, 100 paired seeds; $\pm$ is one standard error.

Observed. For the interleaved variant, sample-path pseudo-regret stayed below the bound $\sqrt{2(d+1)|\mathcal{S}|T \log K}$ in all 100 seeds at every K, that bound running from 2913 to 4228 against realized 455 to 780. The single-copy variant beat action-level weighting at every K, by a margin growing from 19 to 169 as K rose tenfold with $|\mathcal{S}|$ fixed, and beat the interleaved variant throughout. Sweeping the delay at $K = 40$ gave state-pooled 387, 569 and 731 against action-level 653, 699 and 773 at $d = 10$, 50 and 200.

Interpretation. The margin growing in K at fixed $|\mathcal{S}|$ is the prediction that separates the two estimators, so the improvement is attributable to the state channel the effective dimension attributes to it. Interleaving looks like a price of the analysis rather than of the estimator. The narrowing of the gap as the delay grows is what an additive delay term predicts and a multiplicative one does not, which is the diagnostic evidence behind the open problem of Section 6.

Limitation. This is one synthetic family with a fixed drift band, and a simulation cannot establish that a bound holds, only that sample-path pseudo-regret stayed below it in the runs performed.

Study 6. The effective dimension, not $|\mathcal{S}|$, governs the gain. Holding $|\mathcal{S}| = 8$, $K = 40$, $T = 5000$, $d = 20$ and 24 seeds, and varying only the overlap between the rows of P, the measured mean of v_t moves while $|\mathcal{S}|$ does not:

row overlap	mean v_t	state-pooled	action-level	paired gain
0.00	2.719	392.7	599.3	206.6 ± 20.0
0.25	2.013	361.1	528.1	167.0 ± 21.3
0.50	1.532	306.5	406.5	100.0 ± 10.5
0.75	1.168	199.9	229.2	29.3 ± 3.2
0.95	1.009	49.2	50.8	1.6 ± 0.5

Observed. Both the regret and the advantage over action-level weighting fall monotonically with v_t, from a paired gain of 206.6 at $v_t = 2.719$ to 1.6 at $v_t = 1.009$. Nothing about the raw state count changes across the sweep.

Interpretation. $|\mathcal{S}|$ is constant and cannot explain the pattern, while v_t moves with it, which is the comparison Theorem 5 predicts.

Limitation. The two estimators do not become the same object at $v_t \approx 1$. Rather the rows of P agree, so every action induces the same loss and both regrets collapse, making that endpoint uninformative rather than favorable. One parameter of one map family was varied, so the sweep is consistent with the effective dimension governing the gain rather than establishing it.

Study 8. The drift bound describes the right scaling. The construction of Theorem 2 at $K = 40$, $T = 8000$, $h = d$ and 12 paired seeds, sweeping d from 10 to 200, the amplitude ε from 0.02 to 0.20 so that E_2 runs from 12 to 1193, and the drifting coordinates J from 1 to 16. We run four policies: uniform, action-level EXP3, State-EXP3, and the greedy policy on the exact stale vector m_t, the

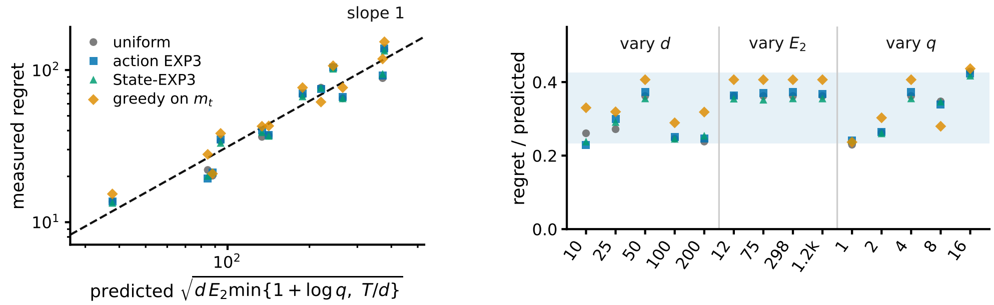

Figure 3: Study 8, the drift channel. **Left,** measured regret against the scale Theorem 2 predicts, over twelve cells in which d varies 20-fold, E_2 100-fold and J 16-fold. The dashed line has slope one. **Right,** the same points divided by that scale, grouped by which parameter the cell sweeps, with the shaded band spanning the observed range. STATE-EXP3 appears here only as one more algorithm on the hard family, where it has no advantage.

last because Theorem 2 claims to survive it. E_2 is measured on the realized instance rather than predicted, so the normalization uses the quantity the theorem uses.

Observed. The predicted scale $\sqrt{d\,E_2 \min\{1 + \log J, T/d\}}$ ranges from 38 to 377 across the grid and the measured regret tracks it (Figure 3). Normalized, the four policies read 0.230 to 0.429, 0.228 to 0.424, 0.238 to 0.417 and 0.237 to 0.437, a spread below 1.9 against a tenfold range in the predictor. The oracle-assisted policy has the largest mean ratio, 0.345 against 0.311, 0.314 and 0.310 for the three that never see m_t.

Interpretation. The three arguments enter the predicted scale differently and the collapse holds in all three sweeps, so it is the form of the bound rather than one fitted constant that tracks the data. That greedy play on the exact m_t sits at the top of the band rather than below it is the concrete content of the claim that the bound survives the oracle.

Limitation. Every cell has $T/d > 1 + \log J$, so only the $1 + \log J$ branch of the minimum is exercised and the T/d saturation branch, which needs d comparable to T, is not probed. A lower bound quantifies over all algorithms and four policies cannot establish that. What these runs check is the scaling of the construction.

Code and environment. The code producing every reported number and figure is at `https://github.com/melikabaghi/state-exp3`. It pins the software environment used for the reported numbers, python 3.14.3 with numpy 2.4.2 and matplotlib 3.10.8, states the seed count for each study, lists one command per numbered study, and commits the stored outputs each figure reads so that every figure rebuilds without rerunning an experiment.

Study 10. Overlap drives the gain, not easier tasks. Study 6 sweeps the overlap and finds the advantage falling as $\widehat{\bar{v}}$ falls, which read at face value contradicts Theorem 5, whose bound improves as $\bar{v}$ falls. That sweep is confounded, since raising the overlap also flattens $c_t = P\theta_t$ across actions, collapsing the comparator gap so that both regrets vanish together. This study holds $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$ and the difficulty fixed, and varies only $\widehat{\bar{v}}$. Difficulty is the uniform-play regret $\sum_t \mathrm{mean}_a c_t(a) - \min_a \sum_t c_t(a)$, a property of the instance with no algorithm in it, and the amplitude of θ is bisected in every cell to hit a common target. The map keeps one contrast direction alive at every overlap and θ is aligned with it, which is what lets the gap survive as the rows agree.

Observed. Across 20 paired seeds the relative gain rises from 11.7 per cent at $\widehat{\bar{v}} = 3.380$ to 19.3 per cent at $\widehat{\bar{v}} = 1.120$, with absolute gains 85.5 ± 8.9 to 106.2 ± 7.9 and a correlation of -0.929 between $\widehat{\bar{v}}$ and the gain. Study 6's unmatched sweep falls from 34.5 to 3.1 per cent over a comparable range (Figure 4).

Interpretation. The two sweeps disagree in sign, and the matched-difficulty sweep agrees with the overlap dependence suggested by Theorem 5, although this experiment uses the single-copy variant rather than the interleaved one that theorem analyses. Overlap is what pooling exploits, and study 6 measures overlap confounded with an instance becoming trivial.

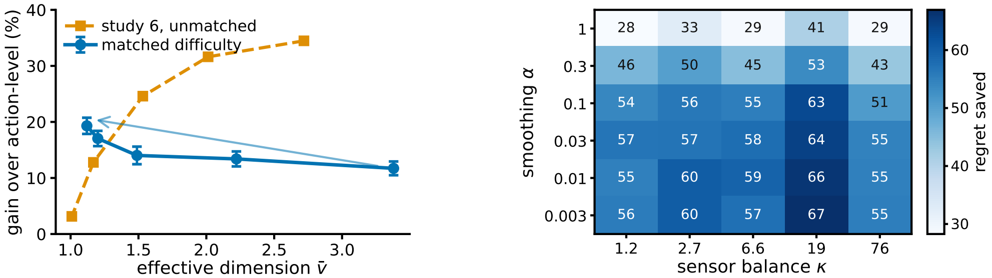

Figure 4: **Left,** study 10. The same overlap sweep with and without holding the task difficulty fixed. Matching the uniform-play regret reverses the trend, supporting the interpretation that the decline in Study 6 is driven by task-difficulty confounding rather than by the pooling mechanism itself. Bars are one standard error of the paired difference. **Right,** regret saved against action-level weighting over the smoothing pseudocount and the sensor balance, 10 paired seeds per cell, from the plug-in sensitivity study whose script is released with the rest.

Limitation. At the two highest overlaps some seeds cannot reach the common target and the realized difficulty is 5 to 13 per cent below it, so the matching is close rather than exact. Since a smaller gap should if anything shrink the advantage, the measured rise is conservative in that direction. The matched gains are also smaller in magnitude than the unmatched ones, which is what fixing the difficulty costs.

Study 10b. The matched-difficulty trend survives a second configuration. Study 10 is a single design point, so the same procedure was re-run at $K = 20$, $|\mathcal{S}| = 12$, $T = 8000$, $d = 20$ and 20 paired seeds, with the common target lowered to 500 so that it is reachable at every overlap. Nothing else changes.

overlap	$\widehat{v}$	R_{unif}	state	action	paired gain
0.00	3.440	500.0	364.7	407.2	42.5 ± 8.2
0.35	2.225	500.0	346.8	401.3	54.6 ± 8.3
0.65	1.518	500.0	309.3	372.4	63.2 ± 6.4
0.85	1.240	493.6	271.8	346.0	74.3 ± 7.2
0.95	1.191	490.1	256.9	327.6	70.7 ± 7.8

Observed. The relative gain rises from 10.4 to 21.6 per cent as $\widehat{v}$ falls from 3.440 to 1.191, and the correlation between $\widehat{v}$ and the absolute gain is -0.972 against study 10's -0.929. Realized difficulty spans 490.1 to 500.0, a 1.020 times spread, so the matching is tighter here than in study 10.

Interpretation. The direction study 10 reports is not an artifact of its single configuration. Two independent design points, differing in K, $|\mathcal{S}|$, T and d, give the same sign and a comparable magnitude.

Limitation. Both configurations use the same construction and the same matching routine, so this tests configuration sensitivity and not construction sensitivity. Both run the practical $m = 1$ variant. An earlier attempt at this replication silently failed because `match_scale` binds its target as a default argument, so the bisection saturated and the difficulty was never matched; the run reported here passes the target explicitly.

Study 14. Many actions resolved by few states is a natural regime. Every study so far draws P from a Dirichlet, which is the assumption under test rather than evidence for it. This one builds an environment to the shape of a recommendation funnel instead. Actions are $K = 200$ catalogue items, states are $|\mathcal{S}| = 6$ ordered engagement depths, and the outcome is a delayed conversion. Three structural facts are imposed because funnels have them, namely that items fall into a modest number of categories sharing an engagement profile, that item quality is heavy-tailed, and that the loss falls with engagement depth and drifts seasonally rather than adversarially. The generated catalogue puts 74 per cent of its mass on no engagement and 1.4 per cent on the deepest.

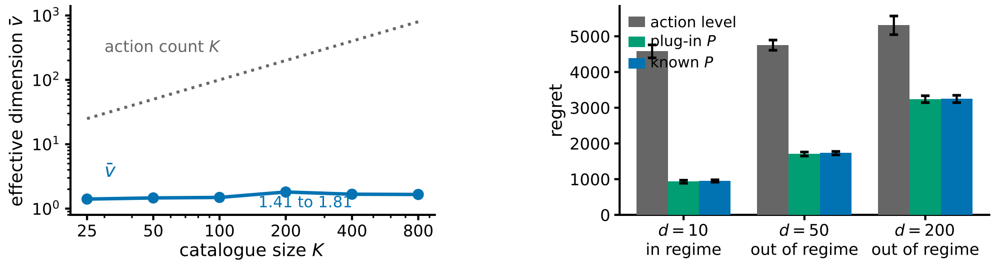

Figure 5: Study 14, a recommendation funnel. **Left,** the effective dimension stays near two while the catalogue grows thirty-twofold, so the gap between K and $\widehat{\bar{v}}$ comes from the catalogue having categories, not from a chosen K. **Right,** the three arms at $T = 20000$ and 8 paired seeds, labelled by whether the regime condition of Section 4 holds. Bars are one standard error. No real data is used anywhere in this study.

Observed. The effective dimension is $\widehat{\bar{v}} = 1.97$ against $K = 200$, a hundredfold gap, and it stays between 1.41 and 1.81 as the catalogue grows from 25 to 800 items (Figure 5). Against action-level weighting the state-pooled variant saves 79.3, 63.6 and 38.9 per cent at $d = 10$, 50 and 200, the largest margins anywhere in this appendix. The plug-in matches known P at every delay, the paired difference never reaching two standard errors. The regime condition $\bar{v}(d+1)\log K \le K + d$ holds at $d = 10$ and fails at $d = 50$ and $d = 200$.

Interpretation. The separation between K and $\widehat{\bar{v}}$ arrives from catalogue structure rather than from a tuned parameter, and it does not close as the catalogue grows, which is the claim the Dirichlet families cannot make. That the condition Theorem 5 needs fails at the delays a funnel has, while the single-copy variant still saves most of the regret, is the same message as study 1 in the setting that motivates the paper. It is evidence for the open problem of Section 6 rather than for the theorem.

Limitation. *No real data is used.* There is no interaction log in this environment, nothing here is fitted to one, and this is not a semi-synthetic benchmark in the usual sense of that phrase. The structure is imposed by hand from qualitative properties, so it shows that the regime is describable and self-consistent, not that any deployed system occupies it. The arms run $m = 1$, covered by Theorem 1, and the environment has no adversarial component, so it exercises the pooling channel and not the drift channel.

Study 16. Pooling still helps against the strongest published algorithm. Every comparison so far is against action-level weighting, EXP3-IX, naive sharing, or the best-state rule. None is the strongest published method for delayed feedback: that is the hybrid Tsallis-plus-entropy FTRL of Zimmert and Seldin (2020), whose delay term is additive rather than multiplicative. We run it on both headline families, with the same environments, seeds and delay convention, under three tunings.

family, cell	State-EXP3	Zimmert and Seldin	same, grid-best	State-EXP3, grid-best
funnel, $d = 10$	949.0	2959.4	2413.8	27.0
funnel, $d = 50$	1728.1	3557.0	2556.4	95.9
funnel, $d = 200$	3244.8	4734.7	2807.7	199.7
overlap 0.00	661.5	731.8	610.4	202.4
overlap 0.65	584.1	673.1	544.0	51.2
overlap 0.95	461.4	540.2	419.4	34.3

Study 16. Regret under three tunings; grid-best columns are chosen by reading these benchmarks.

Observed. With State-EXP3 at the single tuning used everywhere else here and Zimmert and Seldin (2020) at the best of a six-point grid around its own value, State-EXP3 wins every cell, cutting regret by 31.5 to 67.9 per cent on the funnel and 9.6 to 14.6 per cent on the matched sweep. With both at their grid-best, over seven multipliers for Zimmert and Seldin (2020) and six for State-EXP3, State-EXP3 wins every cell again and by more, 66.8 to 98.9 per cent. With Zimmert and

Seldin (2020) at its grid-best and STATE-EXP3 left at its paper tuning, the third column against the first, Zimmert and Seldin (2020) wins the funnel at $d = 200$ and all three matched cells, four of six.

Interpretation. No column is a symmetric protocol. The two grids are finite and unequal, and STATE-EXP3's grid-best sits at its top multiplier in every cell, so its own minimum may lie outside the range searched; the truncation runs against STATE-EXP3. Where it is not held to the worse tuning of the two, the state channel helps against the strongest published method, which is expected, since that method is action-level by construction and cannot read the state at all. Grid tuning helps STATE-EXP3 far more than it helps Zimmert and Seldin (2020), which is consistent with pooling cutting estimator variance and lower variance admitting a larger learning rate.

Limitation. The third column against the first is a real loss and is reported as one. These are our own synthetic families, and a method built for adversarial worst cases is not shown at its best on benign instances. Every grid-best multiplier is chosen by reading these benchmarks, so none of the last three columns is a procedure a practitioner could run; only the first is. Eight paired seeds on the funnel and ten on the matched sweep.

Taken together these studies cover both channels. Studies 1, 6, 10 and 14 probe the state channel and are consistent with the effective dimension, rather than $|\mathcal{S}|$ or K, governing what pooling buys; study 10 had to control for task difficulty before study 6's sweep pointed that way. Study 14 is the only one whose matrix is not drawn from a Dirichlet and reports the largest margins here, on a structure imposed by hand rather than measured. Study 16 places STATE-EXP3 against the strongest published method: ahead in every cell when both are tuned alike, behind in four of six when that method alone is grid-tuned.

F Proof of Theorem 1: additive delay without interleaving

Interleaving cannot give an additive bound. Copy i, which absorbs a share V_i of the budget $V = \sum_i V_i$, has regret $\sqrt{2V_i \log K}$, and Cauchy–Schwarz over the m copies gives $\sqrt{2mV \log K}$, tight when the V_i agree, so the $\sqrt{m}$ in Theorem 5 is not an artifact of the analysis. An additive bound has to come from a single copy.

Throughout, $L_t = \sum_{r \le t-d-1} \hat{c}_r$ and $x_t \propto e^{-\eta L_t}$, so $L_{t+1} - L_t = \hat{c}_{t-d}$, with $\hat{c}_r = 0$ for $r < 1$. The filtration $\mathcal{F}_t$ is the environment's, including outcomes not yet delivered, as in Proposition 4; under it x_t is $\mathcal{F}_{t-d-1}$-measurable, so x_{r+d} is $\mathcal{F}_{r-1}$-measurable.

The drift lemma. The natural route prices the change of measure from x_r to x_{r+d} through the normaliser Z_r, whose reciprocal has no uniformly bounded moment. That is avoidable. The pooled estimate satisfies

$$\langle x_t, \hat{c}_t \rangle = \sum_a x_t(a) \frac{P(S_t \mid a) X_t}{q_t(S_t)} = X_t \le 1$$

identically, on every path, because the denominator is the state marginal of the same distribution that supplies the numerator.

Lemma 12. *If $\eta \le 1/(e(d+1))$ then $x_{t+1}(a) \le (1 + 1/d)\, x_t(a)$ for every t and every a, surely, and hence $x_{t+d}(a) \le e\, x_t(a)$.*

Proof. Strong induction on t. For $t \le d$ nothing is delivered and $x_{t+1} = x_t$. For $t > d$, assume the claim for every $s < t$ and compose it over the d steps from $t - d$ to t, giving $x_t(a) \le (1 + 1/d)^d x_{t-d}(a) \le e\, x_{t-d}(a)$. Contracting with the fixed nonnegative vector $\hat{c}_{t-d}$ and using the identity above, $\langle x_t, \hat{c}_{t-d} \rangle \le e \langle x_{t-d}, \hat{c}_{t-d} \rangle = e X_{t-d} \le e$. With $Z_t = \sum_b x_t(b) e^{-\eta \hat{c}_{t-d}(b)}$ and $e^{-z} \ge 1 - z$, this gives $Z_t \ge 1 - \eta \langle x_t, \hat{c}_{t-d} \rangle \ge 1 - \eta e \ge d/(d+1)$, so $x_{t+1}(a) = x_t(a) e^{-\eta \hat{c}_{t-d}(a)} / Z_t \le x_t(a)/Z_t \le (1 + 1/d) x_t(a)$. □

What the induction bounds is $\langle x_t, \hat{c}_{t-d} \rangle = X_{t-d}\, q_t(S_{t-d}) / q_{t-d}(S_{t-d})$, the state-marginal ratio that the $1/Z_r$ route could not reach. Individual $\hat{c}_r(a)$ remain unbounded; only their x_t-weighted mass is controlled. Cesa-Bianchi et al. (2019) prove the action-level version under $\eta \le 1/(Ke(d+1))$ and Thune et al. (2019) remove the K, using that their estimate is carried by the single action played, so the weight cancels into a same-action ratio. A pooled estimate is spread over every action and does not collapse that way; the identity above replaces the cancellation.

Centring. Let $\zeta_t = \hat{c}_t - X_t\mathbf{1}$, with $\zeta_r = 0$ for $r < 1$. Exponential weights is unchanged by a shift that is constant across actions, so x_t is the same iterate and this is a change of analysis, not of algorithm (Eldowa et al., 2024). Keeping $\gamma(s) = \mathbb{E}[X_t^2 \mid S_t = s]$ inside the state sum,

$$\mathbb{E}\big[\langle x_t, \zeta_t^2\rangle \mid \mathcal{F}_{t-1}\big] = \sum_s \gamma(s)\Big[\frac{\sum_a x_t(a)P(s \mid a)^2}{q_t(s)} - q_t(s)\Big] \le v_t - 1,$$

each bracket being nonnegative by Jensen and $\gamma(s) \le 1$, the final step being Remark 3. Bounding $\langle x_t, \hat{c}_t^2\rangle$ by v_t first and then subtracting does not work: it leaves $v_t - \mathbb{E}[X_t^2]$.

Proof of Theorem 1. Since $\zeta_r(a) \ge -X_r \ge -1$ we have $\eta\zeta_r(a) \ge -\eta$, and Lagrange's remainder gives $e^{-z} \le 1 - z + (e^\eta/2)z^2$ for $z \ge -\eta$. With $\log(1+y) \le y$, the potential $W_t = \sum_a \exp(-\eta\sum_{r\le t-d-1}\zeta_r(a))$ obeys $\log(W_{t+1}/W_t) \le -\eta\langle x_t, \zeta_{t-d}\rangle + (\eta^2 e^\eta/2)\langle x_t, \zeta_{t-d}^2\rangle$. Telescoping from $W_1 = K$ and using $\log W_{T+1} \ge -\eta\sum_{r\le T-d}\zeta_r(a^*)$ for the comparator a^*,

$$\sum_{r=1}^{T-d}\langle x_{r+d}, \zeta_r\rangle \le \sum_{r=1}^{T-d}\zeta_r(a^*) + \frac{\log K}{\eta} + \frac{\eta e^\eta}{2}\sum_{r=1}^{T-d}\langle x_{r+d}, \zeta_r^2\rangle.$$

The shift cancels between the two sides. As x_r is $\mathcal{F}_{r-d-1}$-measurable, $\mathbb{E}[\langle x_r, \hat{c}_r\rangle] = \mathbb{E}[\langle x_r, c_r\rangle]$ and $\mathbb{E}[\hat{c}_r(a^*)] = c_r(a^*)$, and the d rounds undelivered by $T+1$ contribute at most d, so

$$\mathbb{E}[R_T] \le \frac{\log K}{\eta} + \frac{\eta e^\eta}{2}\sum_r \mathbb{E}\big[\langle x_{r+d}, \zeta_r^2\rangle\big] + \sum_r \mathbb{E}\big[\langle x_r - x_{r+d}, \hat{c}_r\rangle\big] + d.$$

Lemma 12 gives $x_{r+d} \le e\,x_r$ surely, and x_{r+d} is $\mathcal{F}_{r-1}$-measurable, so the second sum is at most $e\sum_r \mathbb{E}[v_r - 1] \le eV^-$.

For the third, write $x_{r+d}(a) = x_r(a)e^{-\eta G_r(a)}/Z'$ with $G_r = \sum_{u=r-d}^{r-1}\hat{c}_u \ge 0$ and normaliser $Z' \le 1$, distinct from the one-step Z_t of the Lemma. Then $x_r(a) - x_{r+d}(a) \le x_r(a)(1 - e^{-\eta G_r(a)}) \le \eta x_r(a)G_r(a)$ for every a, trivially where the left side is negative. Since $\hat{c}_r \ge 0$ and x_r, G_r are $\mathcal{F}_{r-1}$-measurable, the conditional expectation is at most $\eta\langle x_r, G_r\rangle$, and $\mathbb{E}[\langle x_r, G_r\rangle] = \sum_{u=r-d}^{r-1}\mathbb{E}[\langle x_r, \hat{c}_u\rangle] \le d$: for each such u, x_r is $\mathcal{F}_{r-d-1}$-measurable and $r-d-1 \le u-1$, so it leaves the conditional expectation and $\mathbb{E}[\langle x_r, \hat{c}_u\rangle] = \langle x_r, c_u\rangle \le 1$. This is measurability, not independence, and at $u = r-d$ the two σ-fields coincide, so the step rests on the timing convention of Section 3. The sum is at most ηdT.

Finally $\eta \le 1/(2e)$ gives $e^{1+\eta} < 3.3$, so $\mathbb{E}[R_T] \le \log K/\eta + \frac{\eta}{2}(3.3V^- + 2dT) + d$. For $f(\eta) = A/\eta + B\eta$ restricted to $\eta \le \eta_{\max}$ one has $\min f \le 2\sqrt{AB} + A/\eta_{\max}$, in the interior and in the binding case alike; with $A = \log K$, $B = (3.3V^- + 2dT)/2$ and $\eta_{\max} = 1/(e(d+1))$ this is the stated bound. $\square$

The cap on η binds only when $d \gtrsim T/(e^2\log K)$, and not at the settings of Appendix E: the funnel study's own rates are $0.0058, 0.0031$ and 0.0016 at $d = 10, 50, 200$ against caps $0.0334, 0.0072$ and 0.0018, so Theorem 1 covers those runs as they were executed. Over three seeds and 20000 rounds the induction has room to spare, $\max_a x_{t+1}(a)/x_t(a)$ reaching 1.006, 1.003 and 1.002 against the permitted 1.100, 1.020 and 1.005.

The two bounds are numerically close on the funnel, where $\bar{v} \approx 2$: the additive form carries $3.3(\bar{v}-1) + 2d$ and the interleaved one $(d+1)\bar{v}$, and these cross near $\bar{v} = 2$. The additive form gains as $\bar{v}$ rises toward $|\mathcal{S}|$, and it describes the algorithm that is actually run, which is the reason to prefer it.